



Basics of Wildlife Health and Management for field Veterinarians



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Topics to be covered in brief

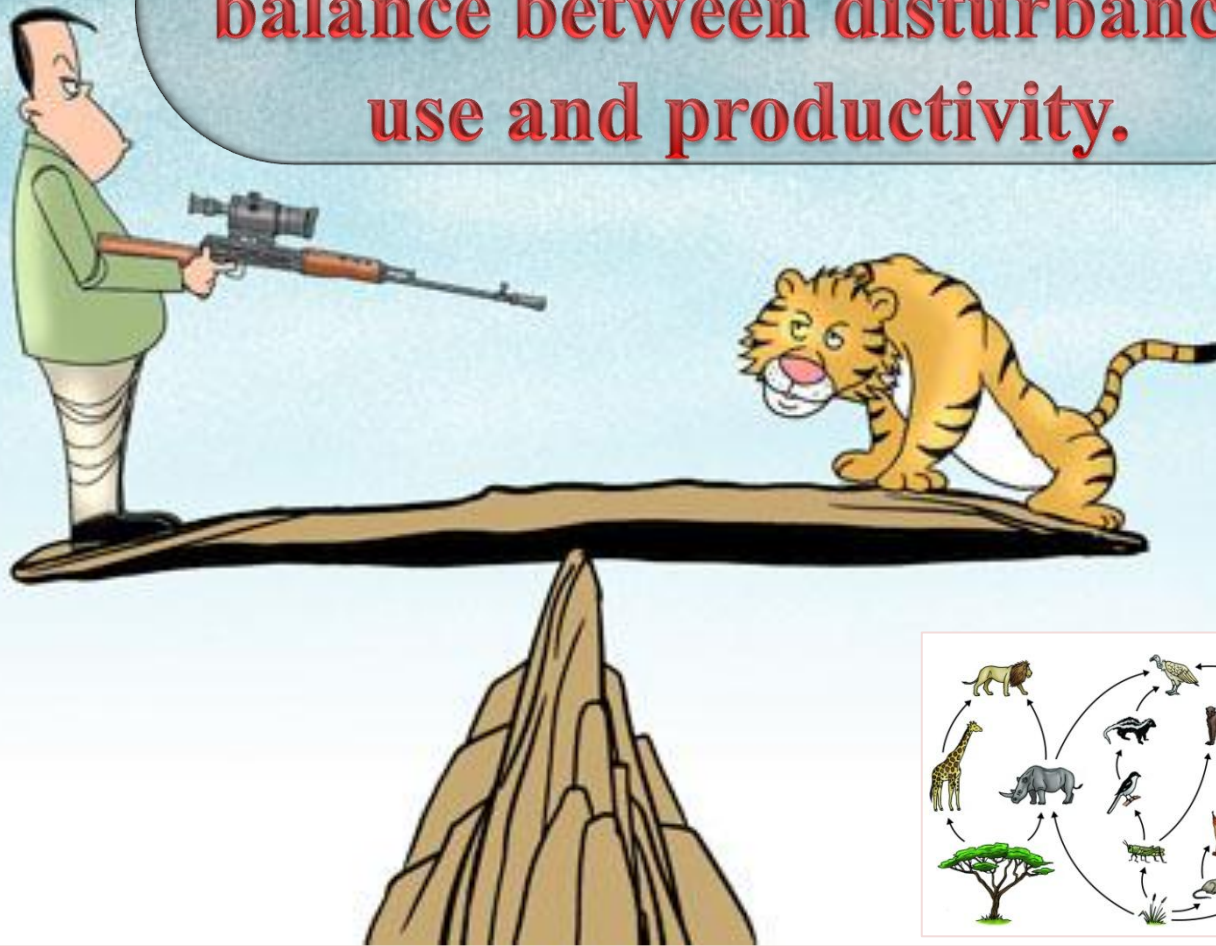
1. Significance of wildlife
2. Zoo animal health challenges in India
3. Diseases and their control measures in wildlife
4. Human wildlife conflict with reference to leopard



Significance of Wildlife in India

Save the human kind !!!

Sustainable Ecosystem is the balance between disturbance, use and productivity.



To save one Tiger is to save many habitats



Lack of foreign revenue



To improve countries market



Zoo animal health challenges in India

Diagnostic uncertainty



Lack of skilled manpower



Emerging diseases



Limited information on standard therapeutic & prophylactic measures



Media and administrative pressure



Managerial issues



Wildlife Health Challenges



What are these suffering from?
They look healthy to me...

Diagnostic uncertainty

- ✓ Brucellosis
- ✓ Early malnutrition
- ✓ Early chronic wasting disease
- ✓ Liver flukes
- ✓ Lungworm
- ✓ Tapeworm
- ✓ Or a host of others!

*Not all Animals, Not all Diseases show
Signs and Symptoms at All times.....*

Administration of Medicaments in wild animals

- I/V – Ear vein



Bear

Jugular vein



Lingual vein



Bear

Medial saphenous vein



Cephalic vein



Elephant



- 1.5 inch 18 G needle
- 2-3 inch 16-18 G



Epidural injection



I/V and S/C – Posterior aspect of ear



Factors affecting ease of medication



1. Personality and training of elephant
2. Handlers confidence and skill
3. Available restraint facilities

Don't underestimate !!!



1. Risk
2. Expertise level
3. Ethical consideration

**Never Believe in
Wild animals**



Reptile medication



Abaxial muscles



Capture and Immobilization



“There is no such thing as Safe anesthetic or safe anesthetic procedure, Just a safe Anaesthetist”.

-Robert Smith

Basic thumb rule

Slow is Fast

Anesthetic dose regimen for important species

Species	Anaesthetic drug	Dose regimen
Tiger	Xylazine and ketamine	1-2 mg/kg and 2-3 mg/kg IM
Lion	Xylazine and ketamine	1-2 mg/kg and 2-3 mg/kg IM
Small felids	Xylazine and ketamine ketamine	0.5 mg/kg and 3-5 mg/kg 5-10 mg IM
Large felids	Medetomidine and ketamine	0.06 – 0.08 mg/kg IM and 2-4 mg IM
Wolf	Xylazine and ketamine	2mg/kg and 10 mg/kg IM
Hyaena	Xylazine and ketamine	0.5-1mg/kg and 8-10 mg/kg IM

Cont...

Species	Anaesthetic drug	Dose regimen
Sloth bear	Xylazine and ketamine	3 mg/kg and 7.5mg/kg IM
	Medetomidin and ketamine	0.05 mg/kg and 3 mg/kg IM
Asian elephant	Xylazine and ketamine Etorphine	0.05-0.1 mg/kg and 0.05-0.1 mg/kg IM 0.0022 mg/kg IM or 1mg/450 kg IM
Spotted deer	Xylazine and Ketamine Medetomidine and ketamine	2-3 mg/kg IM and 2-3 mg/Kg IM 0.05 -1.0 mg/kg and 0.8-3.2 mg/kg IM
Barking deer	Xylazine and ketamine	1.5-2mg/kg IM and 1.5-2 mg/kg IM
Blackbuck	Xylazine and ketamine	1.5-2mg/kg IM and 1.5-2 mg/kg IM



Selected wildlife diseases and disorders

FMD

Fast-Moving-Disease

- ✓ Highly contagious
- ✓ Subsequently, erosions of the oral mucosa and skin of the feet. (Fowler, 1986)

Etiology: Genus -Aphthovirus
of Family *Picornoviridae*



Hosts and Transmission

- ✓ All cloven footed animals

- ✓ Transmission :

 - Ingestion

 - Inhalation



- ✓ Free-ranging elephants are rarely affected when outbreaks occur in ruminants in their territory.

(Kalanidhi ,1992)

- ✓ Wild ruminants –Sylvatic maintenance hosts

Clinical signs and lesions



Management

- ✓ Provide soft feed
- ✓ Foot-slipper sloughs will require prolonged dressings and the use of protective boots.
- ✓ Vaccination may be recommended.

– Ring Vaccination

Trypanosomiasis



Bite of blood
sucking insects
from Local
reservoir host

Ingestion of
infected meat

Bite of blood
sucking insects
from Exotic
animals(Animals
exchange)

Use of dirty
needles



**Possible ways of
Infection in zoo**





Case of Trypanosomosis in wild dog – AAZP Vandalur



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32



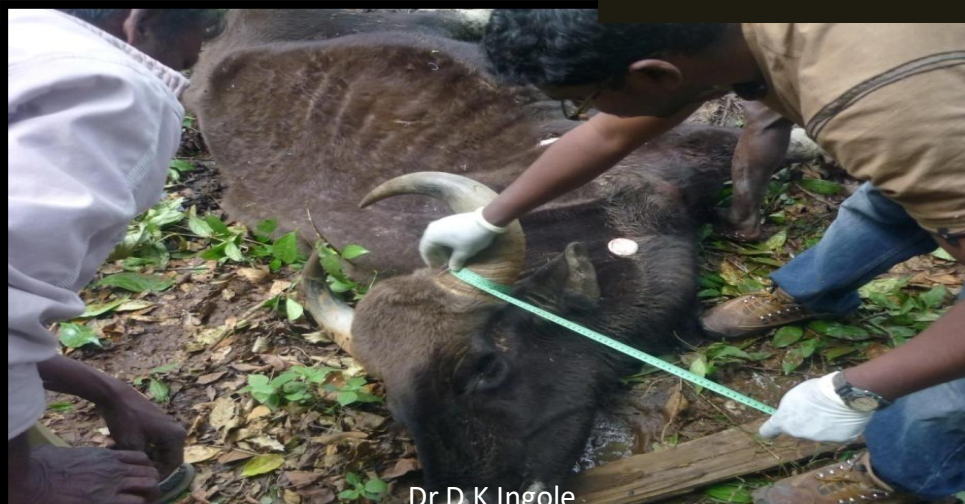
Clinical management

Drug	Dose	Route
Triquin	0.025 mg/kg	S/c
Enrofloxacin	75 mg	I/v
Fluids	300 ml	I/v
Aminoacids (Astymin)	20ml	I/v

In Endemic area, prophylactic treatment with quinapyramine sulphate thrice in a year prevents outbreaks

Tuberculosis

- One of the commonest and fatal zoonoses of wild animals affecting wide range of reptile, avian and mammalian species.
 - In Indian zoos non human primates, cervids, rhinoceros, elephants and felids.
- Host Range:**
- Asian elephants > African elephant
 - Cosmopolitan prevalence.
 - Sex: Female>Male.



Route of infection

- ✓ Aerogenous or inhalation
- ✓ Ingestion
- ✓ Contact transmission
 - Direct
 - Indirect
- ✓ Rarely cutaneous, congenital and mammary
- ✓ Zoonotic disease.



Clinical Signs

- ✓ Weakness
- ✓ Anorexia,
- ✓ Emaciation
- ✓ Low grad fever.
- ✓ Any external abscess below ear, thoracic and abdominal region not responding to treatment is usually looked for TB infection.



Elephants:

- Chronic wasting
- Weight loss
- Coughing/dyspnoea
- Ventral edema
- Exercise intolerance in working elephants.
- In many cases clinical signs are absent.



Diagnosis

- Based on clinical signs and chest X-ray
- **Ante mortem tests**
 - I- Direct Tests
 - Acid fast staining
 - Culture
 - PCR
 - II- Indirect Test
 - Intradermal tuberculin test
 - ELISA
 - Elephant TB stat pak - screening
 - MAPIA (multiantigen print Immunoassay)-confirmatory
 - Gamma interferon

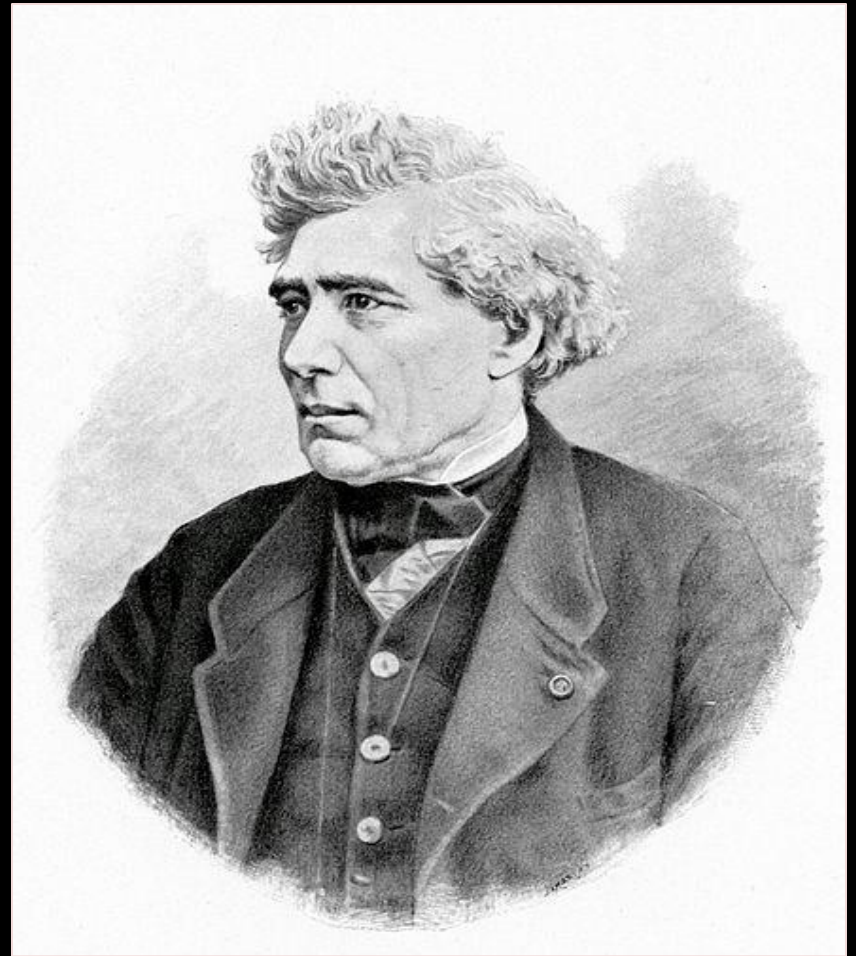
Management with First line of drugs

Starting dosages are (Remember word RISE)

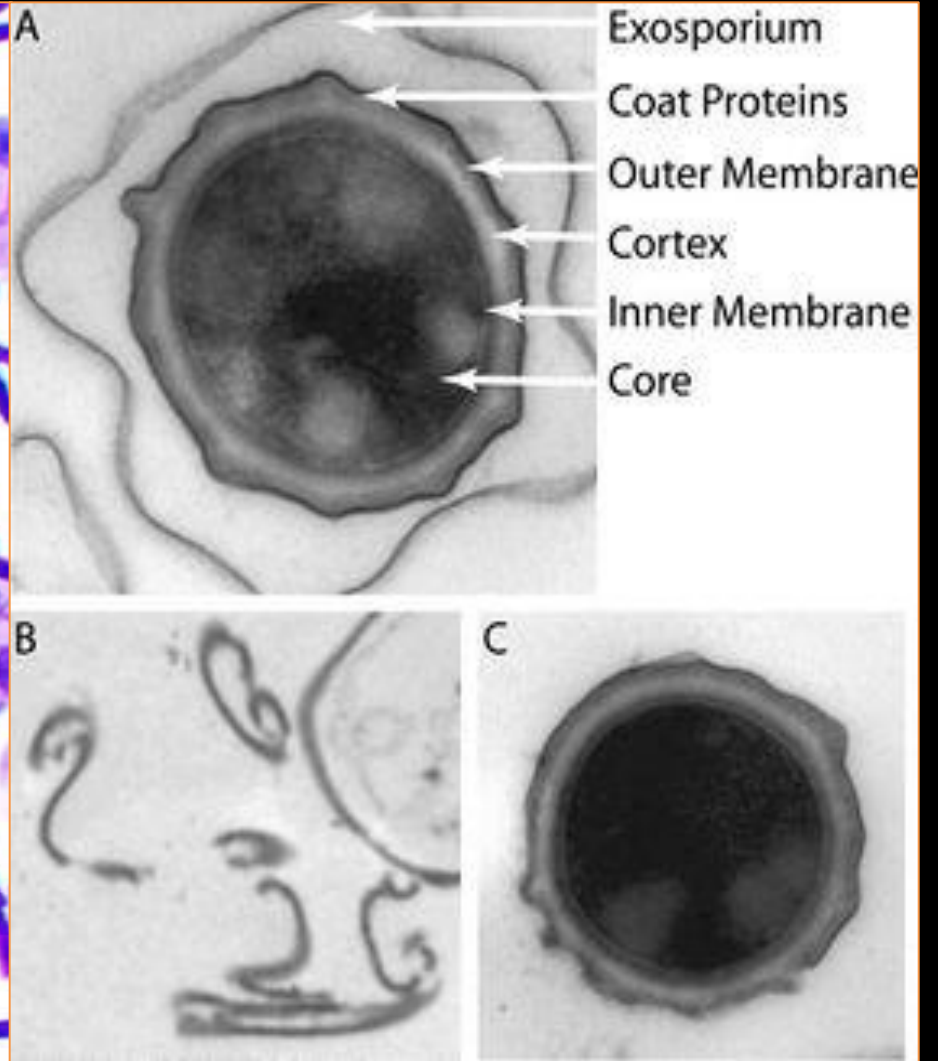
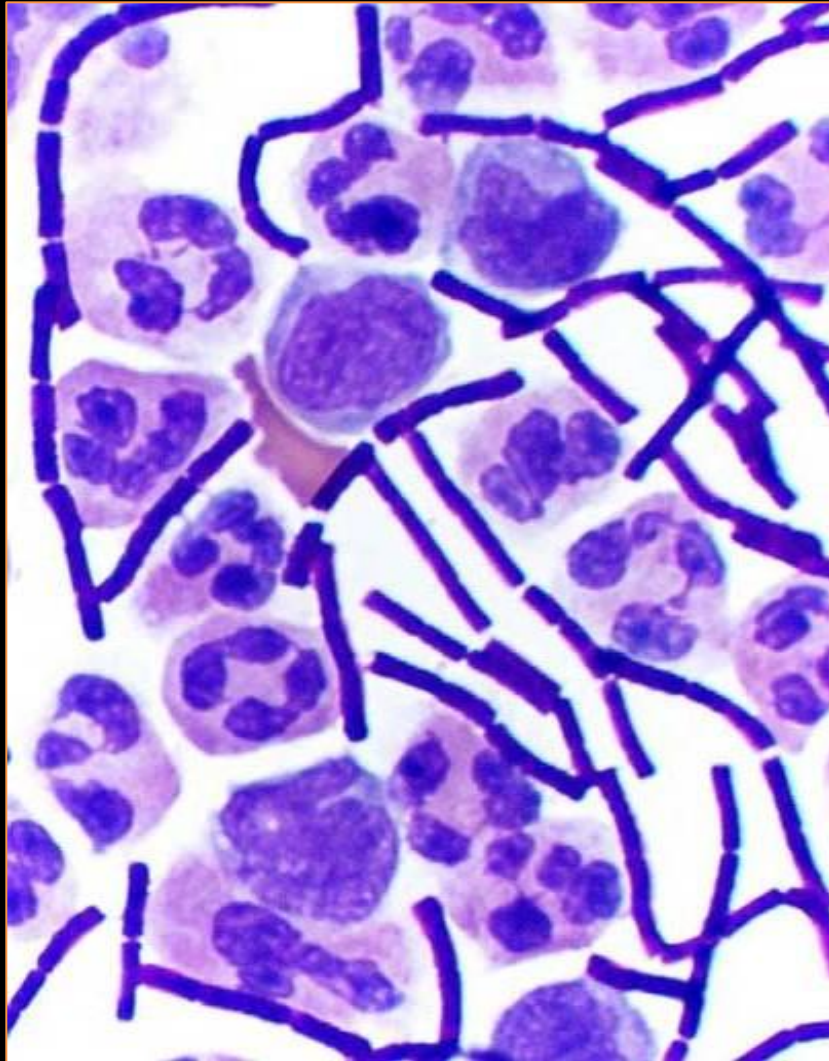
- 5mg/kg for INH Isoniazid (oral or rectal)
- 10 mg/kg for RIF Rifampin (oral)
- 30mg/kg for PZA Pyrazinamide (oral or rectal)
- 30mg/kg for ETH Ethambutol (oral)
- Streptomycin (STR) (Torun, T.,2005)
- After initiation of anti-TB therapy shedding generally ceases within a few weeks.

Anthrax (Siberian plague)

- Oldest documented disease
- Anthrax epidemic –Exodus circa 1491 BC
- Delafond , 1838



Etiology



Host Range

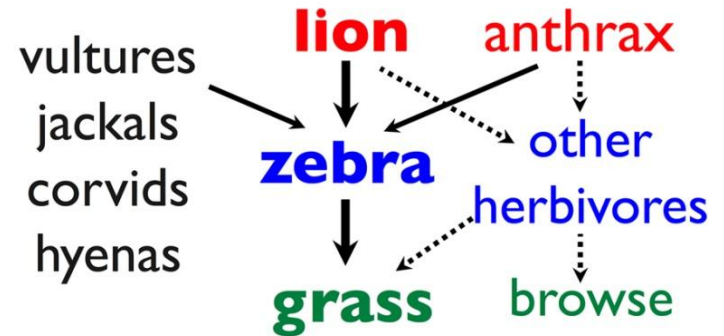


- Wide range of Homeothermic spp
- Herbivores,
- Carnivores
- Birds ???

Do they get infected?



Transmission



Will predators/scavengers get infected by eating Anthrax infected carcass ?



What about in captivity???



Clinical signs

- Frequent micturation
- Hind quarter weakness
- Cutaneous localized swellings
- Edema on inner thigh, testicular and perineal areas in Bison



Diagnosis



Blood smear from the coronary band and hoof lamellae from putrefied carcass

Treatment and Control

- There is no treatment for anthrax in elephants.
- Anthrax Vaccine is a cell-free protein extract of cultures
- **In Elephant:** 2–5 years (1 ml sc)
 - 5–10 years (1.5 ml)
 - 10–15 years (2 ml)
 - 15–20 years (2.5 ml)
 - 20 years and older (3 ml)



Vaccination for Elephant

How to prevent spread in free range?



Brucellosis



Rabies in wild animals



It is a highly fatal disease, mortality being close to 100%.



Rabies is an acute, viral Encephalomyelitis, which affects all warm-blooded animals including humans

Rabies hosts



Animals that don't carry rabies



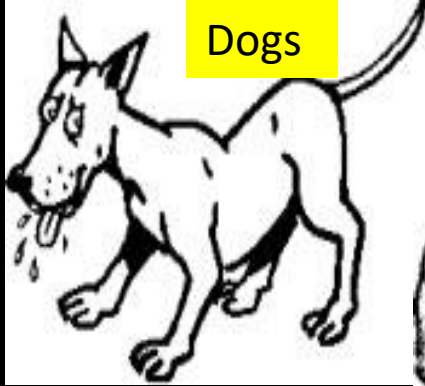
Mammals that are less likely to carry rabies



Rabies in different animals



In India one can get Rabies from these animals



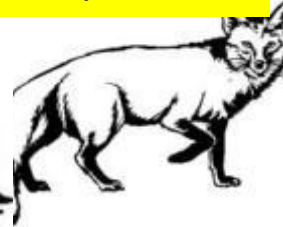
Dogs



Cats



Wolves



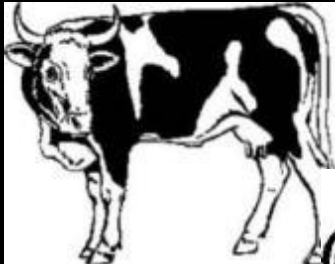
Fox/Jackals



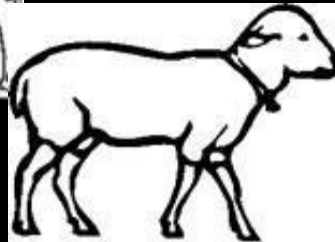
Mongoose



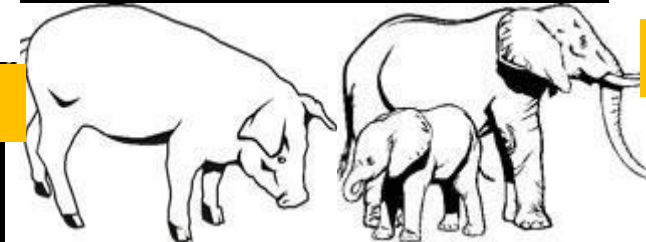
Monkeys



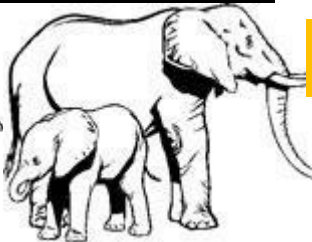
Cattle



Sheep / Goats



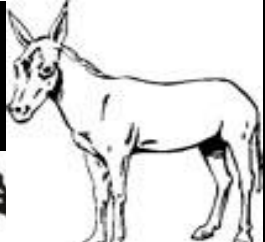
Pigs



Elephants



Horses



Donkeys

Prevention and control of rabies

- Parenteral vaccination
- Use of oral vaccine bait

